Problem F
Six-Seven-Seven-Six

Time limit: 4 seconds

The Council has convened, and your fate is yet unknown.
Solve the riddle laid before you, or never make it home.

N nodes, M edges in sight,
Q rounds, each a pair of nodes, a v and a u,
The distance is the fewest edges between the two.
Is it six-seven-seven-six or less?
The Council wants to know,
Are they close, or is the answer no?

Input

The first line contains three integers N, M and Q.

$$2 \leq N \leq 10^5$$
$$1 \leq M \leq \min(2 \times 10^5, N(N-1)/2)$$
$$1 \leq Q \leq 3 \times 10^5$$

The next $M + Q$ lines each contain two distinct integers u and v, where $1 \leq u, v \leq N$.

The first M lines describe distinct undirected edges. The remaining Q lines describe the queries.

Output

Print a binary string S of exactly Q characters, without spaces. Its i-th character is 1 if the distance between the vertices in query i is at most 6776, and 0 otherwise. Disconnected vertices give 0.

Sample Input 1	Sample Output 1
7 6 3 6 7 2 1 4 7 7 5 3 1 2 3 1 4 6 7 7 6	011